

Udayaditya Palety

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EDUCATION

University of California, Davis | *Bachelor of Science in Biomedical Engineering*

September 2019 - June 2023

Cumulative GPA: 3.6

University of California, San Diego | *Master of Engineering in Bioengineering*

September 2025 - December 2026

Relevant Coursework: Rapid Prototyping, Bioinstrumentation, Biotransport, Statistics, Biomedical Imaging, Circuits Lab

SKILLS

Technical: Machining, Laser Cutting, Soldering, 3D printing (FDM, SLA, and Polyjet), ISO Standardization, FMEA analysis

Software: Solidworks (CAD), MATLAB, Python, Arduino IDE(C++), DICOM, Netfabb, InVesalius, PreForm, LabView

Lab: Gel Electrophoresis, Restriction Digest, MicroPipetting, PCR, Cell Electroporation, Laboratory Maintenance

EXPERIENCE

Process Engineer, Exela Pharma Sciences, LLC

October 2023 - July 2025

- Maintained, set up, and optimized equipment in a cGMP manufacturing environment.
- Created custom apparatus to improve OEE metrics in the aseptic filling suite, increasing efficiency and reducing downtime.
- Designed and fabricated production equipment with SolidWorks to eliminate condensation issues in autoclaved syringe products, improving product quality and compliance.
- Set up and validated pharmaceutical packaging and filling line equipment, while developing SOPs and instructional documentation in compliance with cGMP guidelines.

Undergraduate Researcher, UC Davis Cardiac Signaling Lab, Department of Pharmacology

June 2022 - June 2023

- Performed translational research studying heart disease mechanisms for cardiotoxicity screening and drug R&D.
- Prototyped imaging apparatus through SolidWorks CAD and 3D printing to study cytotoxic drug effects on cardiomyocytes.
- Built an automated microscope assembly line using an Arduino microcontroller for high-throughput sample processing.
- Cataloged laboratory supplies and implemented safety documentation protocols to improve efficiency.

Vice President, B-Hours Engineering Club

January 2020 - June 2023

- Led 60+ students in the B-Hours Engineering Club, facilitating 15+ professional engineering projects.
- Organized 5+ technical workshops teaching soldering, CAD, and laser cutting to 50+ students.
- Directed an 8-member team and faculty partners to develop an interactive parrot engagement device for the San Francisco Zoo Tropical Rainforest Aviary.
- Invented *Mindex*, a pocket-sized viral mitigation device that won first place in the BMES COVID-19 Design Competition.

Project Intern, LV Prasad Eye Institute, Hyderabad India

January 2020 - June 2023

- Utilized MATLAB to streamline the process of custom contact lens design and manufacturing with a team of 4 engineers.
- Created a script to automate spline fitting of spatial data points, extrapolating 3000+ point cloud from 32 clinician inputs
- Operated a high precision CNC manufacturing tool to fabricate contact lenses; reducing development time by 4 hours

PROJECTS

Digital Hand Dynamometer, Biomedical Engineering Senior Design Project

September 2022 - June 2023

- Completed a year-long design project under Dr. Carolyn Patten to develop a digital hand dynamometer to monitor and assess the recovery of stroke patients.
- Conducted user needs surveys and established Engineering Design Standards (EDS) and performance metrics.
- Machined 8+ aluminum components using an end mill and drill press; performed component-level verification testing to ensure ISO compliance.
- Conducted clinical validation tests to demonstrate equivalence to the gold standard market product.

UC Davis Biomedical Engineering Society (BMES), Make-a-Thon Competition

February 2022 - January 2023

- Prototyped two medical devices with a team of four undergraduates for veterinary and NICU applications.
- Awarded *Most Creative Prototype* and *Most Promising Design* in the 2022 and 2023 editions of the competition.
- Manufactured eight prototype components within 24 hours using laser cutting and rapid prototyping techniques.
- Presented final designs to a panel of six industry experts, highlighting design rationale, testing data, and user journey.